



STANDARD OPERATING PROCEDURE
FOR
OPERATION AND MAINTENANCE OF PRETREATMENT PLANT

CLIENT NAME	GINNI FILAMENTS LTD.
LOCATION	PANOLI
SYSTEM	PRETREATMENT SYSTEM
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21

MANUFACTURE AND SUPPLIER

Hydropure Systems Pvt. Ltd.
412, Diamond Industrial Estate,
Dahisar East, Mumbai 400 068, INDIA
Tele Fax: +91 22 289 78725, 2897 9097
www.hydropure.in

R-0	27-08-2021	FIRST DRAFT	N.A.
REV NO.	ISSUE DATE	DESCRIPTION	REMARKS

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 2 OF 27	

TABLE OF CONTENT

1.0	PURPOSE	3
2.0	SCOPE	3
3.0	REFERENCES, ATTACHMENTS AND ANNEXURE	3
3.1	REFERENCE.....	3
3.2	ATTACHMENT.....	3
3.3	ANNEXURE.....	4
4.0	RESPONSIBILITIES	5
5.0	DISTRIBUTION	5
6.0	DEFINITION OF TERMS.....	6
7.0	PROCEDURE	8
8.0	BACKWASH OF MULTI GRADE FILTER (MGF – 101)	13
9.0	REGENERATION SOFTENER (SF – 101 & SF -102)	16
10.0	CIP OF UF SYSTEM	18
11.0	REPLACEMENT OF FILTER CARTRIDGE (CF – 201)	24
12.0	CLEANING OF FILTER CARTRIDGE (CF – 201)	25
13.0	CLEANING OF RO FEED TANK.....	26
14.0	MAINTENANCE	27

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 3 OF 27	

1.0 PURPOSE

To define the procedure for operation and maintenance of the pretreatment plant.

2.0 SCOPE

This SOP is applicable to process pretreatment plant. Details of system are as below

MAKE	CAPACITY	EQUIPMENT CODE	LOCATION
Hydropure Systems Pvt. Ltd.	5600 LPH	—	Pretreatment Water Room

3.0 REFERENCES, ATTACHMENTS AND ANNEXURE

3.1 REFERENCE

- 3.1.1 Pretreatment System Manual
- 3.1.2 Control Panel Operation
- 3.1.3 Control Philosophy for Pretreatment System
- 3.1.4 P & ID Diagram

3.2 ATTACHMENT

- 3.2.1 Attachment–1 Equipment Log Sheets
- 3.2.2 Attachment–2 Chemical Solution Preparation log sheet
- 3.2.3 Attachment–3 MGF Cleaning Records
- 3.2.4 Attachment–4 MGF Depth Checking Records
- 3.2.5 Attachment–5 MGF Media Replacement Records
- 3.2.6 Attachment–6 SF-101 Cleaning (Regeneration) Records
- 3.2.7 Attachment–7 SF-102 Cleaning (Regeneration) Records
- 3.2.8 Attachment–8 UF pre-filter Cleaning Records
- 3.2.9 Attachment–9 UF pre-filter Replacement Records
- 3.2.10 Attachment–10 Tank Cleaning Record
- 3.2.11 Attachment–11 Status Label

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 4 OF 27	

3.3 ANNEXURE

3.3.1 ANNEXURE – 1 SOP of Chemical Solution Preparation

This document elaborated chemical solution preparation procedure and has calculation sheet which states data about inline flow, original chemicals to be used, preparation of desired solution to be dosed, details of dosing pump and setting of flow rates of dosing pumps

Attachment – I of this document gives reference that how operator should generate **‘Monthly Log Sheet - Preparation of Chemical Solution’** to enable him to keep records for this activity

3.3.2 ANNEXURE – 2 Maintenance Schedule sheet

This document elaborates maintenance schedule of major equipments and some of the critical instruments, which need regular precautionary measures apart from calibration

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 5 OF 27	

4.0 RESPONSIBILITIES

4.1 OPERATOR

- 4.1.1** To operate the pretreatment plant as per define procedure
- 4.1.2** To maintain records in log sheets

4.2 SUPERVISOR

- 4.2.1** To ensure that operation of pretreatment plant is carried out as per define procedure
- 4.2.2** To provide training to operator
- 4.2.3** To ensure the cleaning, operation & preventive maintenance of pretreatment plant is carried out as per define procedure
- 4.2.4** To ensure that proper records are maintain

4.3 DEPARTMENT HEAD

- 4.3.1** To ensure that proper operation & maintenance is carried as per SOP
- 4.3.2** To ensure implementation of SOP

4.4 QUALITY ASSURANCE DEPARTMENT

- 4.4.1** To ensure that the job is carried out as per procedure
- 4.4.2** To ensure that proper records are maintain

4.5 QUALITY HEAD

- 4.5.1** To review and approve new or revised SOP's

4.6 FACTORY HEAD

- 4.6.1** To review and approve new or revised SOP's

5.0 DISTRIBUTION

- 5.1** Engineering Department
- 5.2** Quality Assurance Department

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 6 OF 27	

6.0 DEFINITION OF TERMS

6.1	ADV	:	Actuated Diaphragm Valve
6.2	AV	:	Auto Valve
6.3	Bar	:	Bar is a unit of pressure equal to one million (10^6) dynes, equivalent to 10 newtons, per square centimeter
6.4	BFV	:	Butterfly Valve
6.5	BV	:	Ball Valve
6.6	CF	:	Cartridge Filter
6.7	CIP	:	Clean In Place
6.8	CIPFP	:	CIP pump
6.9	CPG	:	Compound Pressure Gauge
6.10	DT	:	Dosing Tank
6.11	DP	:	Dosing Pump
6.12	DV	:	Diaphragm Valve
6.13	FLS	:	Float Type Level Switch
6.14	FS	:	Flow Switch
6.15	GFL	:	Ginni Filaments Ltd.
6.16	HMI	:	Human Machine Interface
6.17	HSPL	:	Hydropure System Pvt. Ltd.
6.18	LLS	:	Low Level Switch
6.19	LPH	:	Litres Per Hour
6.20	LS	:	Level Switch
6.21	MSDS	:	Material Safety Data Sheet
6.22	MWC	:	One of pressure Unit. Meters of Water Column(10.197 MWC is 1 bar pressure)
6.23	M3/hr	:	Unit of volume, Cubic Meter Per Hour
6.24	NRV	:	Non Return Valve
6.25	PG	:	Pressure Gauge
6.26	pH	:	PH is defined as the logarithm of reciprocal of the hydrogen ion concentration or is equal to the logarithm of the hydrogen ion concentration with negative sign $PH = \log_{10} 1/(H^+) = -\log_{10} (H^+)$

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 7 OF 27	

6.27	PID	:	Piping and Instrumentation Diagram
6.28	PLC	:	Programmable Logic Controller
6.29	PRE	:	Pre-treatment System
6.30	PT	:	Pressure Transmitter
6.31	PS	:	Pressure Switch
6.32	PSIG	:	PSIG is a unit of pressure as pounds per square inch gauge
6.33	PU	:	Polyurethane
6.34	RECP	:	Recirculation Pump
6.35	RM	:	Rotameter
6.36	RWT	:	Raw Water Tank
6.37	SDI	:	Silt Density Index
6.38	UB	:	'U' Shaped Bend
6.39	UPVC	:	Unplasticized polyvinyl chloride
6.40	VF	:	Vent Filter
6.41	QA	:	Quality Assurance

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 8 OF 27	

7.0 PROCEDURE

7.1 SAFETY PRECAUTIONS

- 7.1.1** Read 'Operating and Maintenance Manual' of individual equipment carefully before proceeding with operation of pretreatment system. Safety precautions specified in these manuals shall be followed carefully. Do not deviate from the data specified in these manuals
- 7.1.2** While using any cleaning chemical, MSDS of respective chemical shall be handy before using it. Read its handling and safety procedures carefully before handling them
- 7.1.3** Ensure that all cleaning chemicals are dissolved and well mixed before circulating the cleaning solution through the system
- 7.1.4** Refer 'Dosing Solution Preparation' SOP while preparing online dosing chemical solutions (ANNEXURE – 1)
- 7.1.5** Wear hand gloves and face mask while handling chemicals
- 7.1.6** Follow safety processes published by respective EHS department

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 9 OF 27	

7.2 MODE OF OPERATIONS

There are two service modes in the system as follows

- 7.2.1 AUTO MODE** – It is always recommended to operate the system in ‘**AUTO MODE**’ through PLC control panel
- 7.2.2 MAINTENANCE/MANUAL MODE** – Means manual operation of plant. It is advisable to use ‘**MAINTENANCE MODE**’ only during maintenance or incase of failure or replacement of any equipment or component. Proper records of these activities shall be maintained in respective sheet.

These modes are selectable from HMI and at a time only one mode can be executed. User has to touch ‘Maintenance Mode’ icon on HMI to select the Maintenance mode and same way ‘Auto Mode’ icon to select auto mode operation. If user has selected the Maintenance mode, then ‘Auto Mode’ selection icon will be disabled and will become enabled only when ‘Maintenance Mode’ icon is de-selected and vice a versa.

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 10 OF 27	

7.3 POSITION OF VALVES

Following procedures are to be followed during working of the system

7.3.1 Position of Manual Valves – Position of manual valves recommended in below table will remain same for 'Auto' as well as 'Manual' mode of operations

Refer below table for the position of manual valves of pretreatment system-



Position of manual valves specified in below table refers to service mode (either in AUTO or MANUAL mode) i.e. operation of pretreatment for the production of treated water to suffice downstream requirement

Position of some of these valves may change during backwash of Multi Grade Filter. The same will be specified in respective section



DO NOT FORGET TO CHANGE position of respective downstream valves of equipment during cleaning operation. OTHERWISE it may cross contaminate downstream system

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE (DURING SERVICE MODE)
A	SERVICE MODE OF RAW WATER AREA		
1.	DV-101	At Suction Line of RECP-101	Open
2.	DV-102	At Discharge Line of RECP-101	Open Partially*
3.	NV-101	At Recirculation line in Raw Water Tank	Open
B	SERVICE MODE OF MULTI GRADE FILTER (MGF-101)		
4.	DV-103	At Suction Line of RWP-101	Open
5.	DV-104	At Discharge Line of RWP-101	Open Partially*
6.	NV-102	Isolation valve of DSP-101	Open
7.	NV-103	Isolation valve of DSP-101	Open

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 11 OF 27	

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE (DURING SERVICE MODE)
C	BACKWASH MODE OF UF SYSTEM		
8.	DV-105	Suction Valve of BFP-101	Open
9.	DV-106	Discharge Valve of BFP-101	Open Partially*
E	SAMPLING VALVES OF COMPLETE SYSTEM		
10.	Sampling valves shall be closed throughout the operation (Service mode or cleaning) of system. Sampling valves shall only be opened for collection sampling		



* This setting will be done by Supplier's personnel during commissioning and shall not be changed unless and until there is change in feed water quality or downstream demand. Contact supplier before making any changes

Discharge valve of any pump shall be opened partially before starting system if this setting needs to be changed. Then start the system (Auto Mode or Manual Mode) and adjust this valve to get the desired recommended flow to avoid any damage to downstream equipment

7.3.2 Position of Auto Valves –

7.3.2.1 In '**Auto Mode**' - operation of auto valves will be taken care by PLC control panel

7.3.2.2 In '**Manual Mode**' - auto valves of pretreatment system shall be operated manually through HMI screens. Sequence of this operation shall be listed in 'Operation of system in Manual Mode' section of this document

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 12 OF 27	

7.4 CHEMICAL SOLUTIONS FOR ONLINE DOSING

7.4.1 REFER –

7.4.1.1 ANNEXURE–1, 'SOP of Chemical Solution Preparation' to prepare chemical solution

7.4.1.2 ATTACHMENT-1 for 'Chemical Solution Preparation log sheet'

7.4.2 Refer ANNEXURE-2 of SOP of Chemical Solution Preparation' to verify settings of dosing pumps, concentration of chemical solution to be prepared

7.4.3 Online dosing chemicals shall be prepared daily

7.4.4 Discard last day's left over chemical and clean dosing tanks properly before adding freshly prepared solution

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 13 OF 27	

8.0 BACKWASH OF MULTI GRADE FILTER (MGF – 101)

The water enters sand filter from top and gets filtered through carefully selected graded sand. The suspended solids are trapped / arrested by the sand media and these are removed /drained during backwash of the sand filter. The sand filter is recommended to be backwashed once a day or on attaining a differential pressure of above 0.5 kg/cm²

- 8.1 BACKWASH FREQUENCY:** Once in a day OR when differential pressure is more than 0.5 kg/cm² or whichever is earlier



Backwash of MGF-101 shall be done manually in 'MANUAL' MODE
Follow sequence of valves properly as specified below for backwash in manual mode

8.2 PROCEDURE:

- 8.2.1 STOP** Raw Water Pump, RWP-101 if in 'MANUAL' Mode

- 8.2.2** And follow below steps to backwash of MGF-101 unit



Operate upstream and downstream system, Raw Water inlet & Pretreatment system in service mode.

8.2.3 BACKWASH:

- 8.2.3.1** System will perform backwash at that time system have to change the position of all associated manual valves as follows –

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE
A	BACKWASH OF MULTI GRADE SAND FILTER (MGF – 101)		
1.	ADV-104	At Backwash Inlet Valve of MGF-101	Open
2.	ADV-105	At Backwash Outlet Valve of MGF-101	Open
3.	ADV-102	At Service Inlet Valve of MGF-101	Close
4.	ADV-103	At Service Outlet Valve of MGF-101	Close
5.	ADV-106	At Drain Valve of MGF-101	Close

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 14 OF 27	

SR. NO.	MANUAL VALVE TAG NO.	DESCRIPTION OF VALVE	POSITION OF MANUAL VALVE
6.	ADV-107	Air Release Valve of MGF-101	Open initially during Back flush Cycle
B	FORWARD FLUSH OF MULTI GRADE SAND FILTER (MGF – 101)		
7.	ADV –102	Inlet Valve of MGF-101	Open
8.	ADV-106	At Drain Valve of MGF-101	Open



Position of which valves being changing from service mode to backwash of MGF has been highlighted in above table for ready reference

Do not start backwash if water level in raw water tank is BELOW 'Low' level. Continue filling tank till water level goes ABOVE 'Low' and then initiate backwash

- 8.2.3.2 Backwash will be in upward direction so air shall be released through air release valve. **CLOSE** air release valve, **ADV – 107**. once **water start flowing continuously through it**
- 8.2.3.3 Backwash MGF – 101 for about 15-20 minutes
- 8.2.3.4 Then **STOP** selected Raw Water Pump, **RWP-101**. And put system in forward flush mode

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 15 OF 27	

8.2.4 FORWARD FLUSH / RINSE CYCLE:

- 8.2.4.1 **Change the position of valves** as specified in above table to initiate forward flush of system. Forward flush is carried out to remove any leftover of backwash on surface of media otherwise it may get carried away to downstream system
- 8.2.4.2 **FORWARD FLUSH: START** selected Raw Water Pump, **RWP-101**.
- 8.2.4.3 **Continue flushing MGF for 5-10 minutes or until filtrate is visibly clear**
- 8.2.4.4 Then **STOP** selected Raw Water Pump, **RWP-101**. And put the system in service mode

8.2.5 SERVICE MODE:

- 8.2.5.1 Change the position of valves as specified in section no. 8.2.1 of this document to **PUT** MGF in **SERVICE MODE**
- 8.2.5.2 Then **START** selected Raw Water Pump, **RWP-101**. And **PUT** the system in **SERVICE MODE**

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 16 OF 27	

9.0 REGENERATION SOFTENER (SF – 101 & SF -102)

The resins of the softener are needed the regeneration process, as the service cycle of the softener is complete it has to go manually in regeneration mode

9.1 REGENERATION FREQUENCY: AS PRE DEFINE OBR (OUTPUT BETWEEN REGENERATION)



Backwash of SF-101 & SF-102 shall be done manually in 'MANUAL' MODE
Follow sequence of valves properly as specified below for softener in manual mode

9.2 PROCEDURE:

9.2.1 STOP Raw Water Pump, RWP-101. if in 'MANUAL' Mode

9.2.2 BRINE INJECTION CYCLE:

9.2.2.1 Before going for brine injection cycle make sure that the level of Brine Tank is at its required level

9.2.2.2 **Open** ADV-111, ADV-112 and ADV-113 to perform the brine injection cycle if SF-101 will be working. OR **Open** ADV-120, ADV-121 and ADV-122 to perform the brine injection cycle if SF-102 will be working.

9.2.2.3 Perform the cycle for **20 – 25 minutes**

9.2.3 FAST RINSE CYCLE

9.2.3.1 **Open** ADV-109 and ADV-114 to perform the fast rinse cycle if SF-101 will be working. OR **Open** ADV-118 and ADV-123 to perform the slow rinse cycle if SF-102 will be working.

9.2.3.2 Perform the cycle for **5 – 10 minutes**

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 17 OF 27	

9.2.4 SLOW RINSE CYCLE

9.2.4.1 **Open** ADV-112 and ADV-113 to perform the slow rinse cycle if SF-101 will be working. OR **Open** ADV-121 and ADV-122 to perform the slow rinse cycle if SF-102 will be working.

9.2.4.2 Perform the cycle for **10 – 15 minutes**

9.2.5 BRINE TANK FILLING CYCLE

9.2.5.1 **Open** ADV-109 and ADV-111 to perform brine tank filling cycle if SF-101 will be working. OR **Open** ADV-118 and ADV-120 to perform the brine tank filling cycle if SF-102 will be working.

9.2.5.2 Then **STOP** Raw Water Pump, **RWP-101**. And put the system in service mode

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 18 OF 27	

10.0 CIP OF UF SYSTEM

The surface of Ultrafiltration (UF) membranes is subject- to fouling by foreign materials that may be present in the feed water, such as suspended solids, colloids, organics and biological matter. The term “fouling” includes the build-up of various kinds of layers on the membrane surface, including scaling. Following precautions has been taken during design stage to protect UF membranes from fouling or scaling

- ❖ **PRETREATMENT** of the feed water prior to the UF process is designed to reduce contamination of the membrane surfaces. This is accomplished by installation of sufficiently designed filters, oxidation processes etc. for challenging feed water application
- ❖ Incorporation of **PERIODIC BACKWASH** to remove loosely associate particle from membrane surface so that they should not harden over there causing fouling or scaling of membrane

In spite of all these precautions, there are some unavoidable reasons like seasonal variation in feed water quality, slow built up of contaminants etc. lead to fouling / scaling of UF membranes. The fouling of membrane surfaces results in the gradual performance decline of the UF System in terms of lower sustainable filtrate flow or flux, high operating pressure drop and/or high chemical and power consumption. Hence UF system performance shall be monitored on a regular and frequent basis. If cleaning is delayed too long, fouling may become irreversible and result in potential physical damage to the UF module or downstream equipment

10.1 CRITERIA FOR CIP OF UF SYSTEM:

10.1.1 Ultrafiltration system shall be cleaned to restore membrane permeability, when one or more of the below parameters are encountered during regular operation:

10.1.1.1 **FILTRATE** flow **DROPS** by **20%**

10.1.1.2 Pressure Drop (ΔP) across the membrane exceeds **1 bar** (g) (14.5 psig)

10.2 CIP FREQUENCY:

10.2.1 6 months $\pm$ 1 week OR above mentioned situation in spite of regular backwash, whichever is earlier

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 19 OF 27	

10.3 CIP CHEMICALS & CYCLES PREFERRED:

10.3.1 Below mentioned table gives the chemicals preferred with the time for CIP cycles.

10.3.1.1 CIP-1 is to be performed when the filtrate flow drops by 20% & Pressure drop (ΔP) across the membrane exceeds 1 **bar** (g) (14.5 psig) or at regular 6 month CIP cycle as mentioned above

10.3.1.2 CIP-2 is to be performed if the pressure across the UF is not get recovered after performing CIP-1 as specified above

10.3.1.3 **Schedule CIP as per point no. 10.2.1 shall be performed with CIP-1 chemicals only**

SR. NO.	CHEMICAL	CHEMICAL.	CONCENTRATION /pH	QUANTITY REQUIRED
1	CIP-1	hypochlorite NaOCl	200 ppm	Total solution - 400 liters (Reference) Chemical (6% NaOCl) – 1.4 liters Water – 398.6 liters
2		Sodium Hydroxide NaOH	Add to adjust pH up to 12	Add slowly NaOH solution to bring pH to 12
3	CIP-2	Citric Acid $C_6H_8O_7$ OR Hydrochloric Acid HCl (0.2%)	Citric Acid - 2% of pH – 2 OR HCl – 0.2%	Total solution - 400 liters Check pH. It should be 2



IF PRESSURE DROP IS NOT GET RECOVERED AFTER PERFORMING BOTH THE CIP CYCLES (CIP-1 & CIP-2) THEN CONTACT HYDROPURE SYSTEMS PVT. LTD.

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 20 OF 27	

10.4 CIP PROCEDURE:

10.4.1 MODE SELECTION – Select the 'MANUAL MODE' from HMI before starting the CIP cycle

10.4.2 REGULAR BACKWASH – perform the backwash of the UF system as specified in above sections having UF air flushing, air release, UF backwash Up-Rinse, Down-Rinse & fast flush cycles

10.4.3 READINESS FOR CIP – After completion of backwash of the UF system, **DE-SELECT** 'MANUAL MODE'. Now system is in 'IDLE / STANDBY MODE'. Ensure following positions of valves and pumps before starting CIP

SR. NO.	VALVE/PUMP TAG NO.	DESCRIPTION OF VALVE/PUMP	POSITION OF AUTO VALVE/PUMP
1.	AV-101, AV-102, AV-103, AV-104, AV-105, AV-106	Auto valves of UF system	Closed
2.	ROFP-101	RO Feed Pump	START

10.4.4 CIP SOLUTION PREPARATION -

10.4.4.1 Use RO Feed Tank to prepare chemical solution

10.4.4.2 Refer table specified in point no. 10.3.1.3 to prepare CIP solution

10.4.5 SYSTEM USED -

10.4.5.1 **RO FEED TANK i.e. T-201**

10.4.5.2 **RO FEED PUMP i.e. ROFP-201**

10.4.6 CIP SYSTEM CONNECTION –

10.4.6.1 Install new cartridge filter in filter housing, CF – 101

10.4.6.2 Connect a one end of hose pipe before RO High Pressure Pump (CIP-101 Outlet) and connect second end of hose pipe at point provided before UF module with the help of hose pipe. Similarly connect the CIP outlets (CIP –

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 21 OF 27	

103 outlet & CIP – 104 outlets) to the RO Feed Tank inlets using hose pipes. Ensure that all hose pipes are properly connected

10.4.7 Below is the sequence of operation which provides general valve sequence for the steps of the CIP procedure. A description of each step of the CIP procedure is provided in this section

10.4.7.1 RECIRCULATION 1 –

- Circulate the selected CIP solution through the UF system for **45 minutes**. Throughout the recirculation check the concentration of the CIP solution from the RO Feed Tank After every 15 minutes, if the concentration drops below the specified limit as per above table then top up the solution in the tank



Ensure that concentration & pH of the CIP Solution is as per specified in the table of point no. 10.3.1.3

10.4.7.2 SOAKING –

- After completion of the 45 min recirculation – 1, switch off the RO Feed Pump & keep the entire system in **SOAKING MODE** for another 45 minutes

10.4.7.3 RECIRCULATION 2 –

- After completion of soaking cycle, start the RO Feed Pump and keep system under recirculation for 45 minutes, same as of RECIRCULATION – 1 as mentioned above

10.4.7.4 FLUSHING 1 –

- STOP RO Feed Pump, ROFP – 201, remove hoses which are connected to CIP ports, CIP – 102, CIP – 103 and CIP – 104 from tank and put in nearest drain point. Then start Back Flush Pump till ‘low level’ of RO Feed Tank, T – 201. At low level Back flush pump will get tripped. **Do not change the position of hoses**

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 22 OF 27	

- **RE-FILL** RO Feed Tank, T –201, with UF permeate water. **RE-START** RO Feed Pump to flush UF system again. Continue flushing till ‘Low Level’ of T – 201

10.4.7.5 **DISMANTLING OF CIP –**

- Remove hoses connected at CIP ports, CIP – 101, CIP – 102, CIP – 103 and CIP – 104 from UF system
- Handle CIP system separately for cleaning after completion of UF flushing

10.4.7.6 **FLUSHING 2 –**

- Perform the forward flush of UF system as specified in section 10.3
- Continue this forward flush of UF system till pH of water at **UF outlet is same as of Feed water**
- After completion of forward flush cycle the CIP of the system completed

10.4.7.7 **RETURN TO SERVICE MODE –** Now UF system is ready to put in service mode

Start UF system in service mode and monitor the system for next 4 – 5 hours. During this period water can be used for downstream requirement. If differential pressure across the system is <1 bar then continue with service mode. IF NOT then perform CIP – 2

10.4.7.8 **CLEANING OF CIP SYSTEM**

- **Manually drain RO Feed Tank by opening tank drain valve**
- **Wipe out remaining solution from tank with soft cloth**
- **Discard used cartridge from CF – 201**

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 23 OF 27	

10.5 CLEAN-IN-PLACE (CIP) TIPS

- 10.5.1** Always measure the pH during cleaning. If the pH increases more than 1 pH units during acid cleaning, more acid needs to be added. If the pH decreases more than 1 pH units during alkaline cleaning, more caustic needs to be added
- 10.5.2** Inspect the appearance of the spent CIP solution at various points during the recycle and soak steps. The visual appearance sometimes provides indication of the type of fouling, especially in the case of inorganic metals
- 10.5.3** Fresh cleaning solution needs to be prepared when the cleaning solution becomes turbid and/or discoloured. The cleaning is repeated with a fresh cleaning solution
- 10.5.4** Use least harsh cleaning solution possible, including cleaning parameters of pH and solution strength

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 24 OF 27	

11.0 REPLACEMENT OF FILTER CARTRIDGE (CF – 201)

11.1 CRITERIA FOR REPLACEMENT OF CARTRIDGE FILTER:

11.1.1 If pressure difference (ΔP) across cartridge filter increases above **0.7 Kg/cm²**

11.2 REPLACEMENT FREQUENCY:

11.2.1 1 Month $\pm$ 1 week or above mentioned criteria, whichever is earlier

11.3 PROCEDURE:

11.3.1 **STOP** 'PRETREATMENT SYSTEM' if system is operational in '**AUTO**' Mode or STOP RO Feed Pump, **ROFP-201**. if in '**MANUAL**' Mode

11.3.2 After Ensuring that the pressure in the cartridge filter is zero (check in PG – 201 & 202), dismantle piping from outlet nozzle

11.3.3 Open the cartridge filter cover through TC clamp which is holding body cover with base.

11.3.4 Remove the existing filter elements and clean them

11.3.5 Observed the particulates to see, if there is any abnormal carryover of sand or any other material

11.3.6 Drain the cartridge filter body and wash it with water

11.3.7 Install new filter cartridge elements and prefix the covers tightening TC clamp

11.3.8 Connect upstream piping to outlet nozzle

11.3.9 Meanwhile check pressure across the cartridge filter (CF – 201) by checking inlet (PG – 201) and outlet pressure (PG – 202). If differential pressure is **< 0.7 Kg/cm²** , then consider that installed cartridges are working properly

11.3.10 Check for any visual leakages. If not that means piping is connected properly

11.3.11 Now system is ready to operate in 'AUTO' mode or 'MANUAL' mode as desired.



Record replacement of cartridge filters in equipment log sheet

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 25 OF 27	

12.0 CLEANING OF FILTER CARTRIDGE (CF – 201)



If cartridge for replacement is not available in market then cleaning procedure to be perform.

12.1 CLEANING FREQUENCY:

12.1.1 After 1 month $\pm$ 7 days

12.2 PROCEDURE:

12.2.1 **STOP** 'PRETREATMENT SYSTEM' if system is operational in '**AUTO**' Mode or STOP RO Feed Pump, **ROFP-201**. if in '**MANUAL**' Mode

12.2.2 After Ensuring that the pressure in the cartridge filter is zero (check in PG – 201 & 202), dismantle piping from outlet nozzle

12.2.3 Open the cartridge filter cover through TC clamp which is holding body cover with base.

12.2.4 Remove the existing filter elements

12.2.5 Clean them with UF permeate water from reverse direction to normal flow

12.2.6 Observed the particulates to see, if there is any abnormal carryover of sand or any other material

12.2.7 Drain the cartridge filter body and wash it with water

12.2.8 Install cleaned cartridge elements back into housing and prefix the covers tightening TC clamp

12.2.9 Connect upstream piping to outlet nozzle

12.2.10 Meanwhile check pressure across the cartridge filter (CF – 201) by checking inlet (PG – 201) and outlet pressure (PG – 202). If differential pressure is as less than 0.7 Kg/cm², then consider that installed cartridges are working properly

12.2.11 Check for any visual leakages. If not that means piping is connected properly

12.2.12 Now system is ready to operate in 'AUTO' mode or 'MANUAL' mode as desired.



Record cleaning of cartridge filters in equipment log sheet

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 26 OF 27	

13.0 CLEANING OF RO FEED TANK

13.1 CLEANING FREQUENCY:

6 months $\pm$ 1 week

13.2 PROCEDURE:

13.2.1 Close outlet valve of the tanks

13.2.2 Check the water level of the tank

13.2.3 Add **sodium hypochlorite (6% w/v concentration - approx.)** in the tank manually & stir with the help of the stainless steel ladder to make the solution of approx. 50 ppm

13.2.4 Hold on this condition for minimum 5-6 hrs or overnight if possible

13.2.5 Check the concentration of the solution. Top up with chemical if concentration drops down below 50 ppm during hold time after every 2 hour

13.2.6 After completion of holding period drain out the water from tank completely using drain valve

13.2.7 Fill the tank with fresh water up to 1/3rd level. Stir this water with Stainless steel rod wounded with the clean soft cloth at the rod tip. Ensure that it cleans the bottom of tank properly

13.2.8 Rub the bottom surface of the tank using this rod & drain out the water from the tank completely

13.2.9 Repeat the process specified above till the bottom surface of the tank get cleaned from the chemical deposition

CLIENT	GINNI FILAMENTS LTD.		
DOCUMENT NAME	STANDARD OPERATING PROCEDURE OF PRETREATMENT SYSTEM	REV NO – 00	
DOCUMENT NO.	HSPL/GFL/PR681/SOP/PRE/21	PAGE 27 OF 27	

14.0 MAINTENANCE

Refer **ANNEXURE – 2** 'Maintenance Schedule' for preventive maintenance of major equipments and critical instruments.

Every Breakdown of Equipment should be recorded in Equipment History Card.